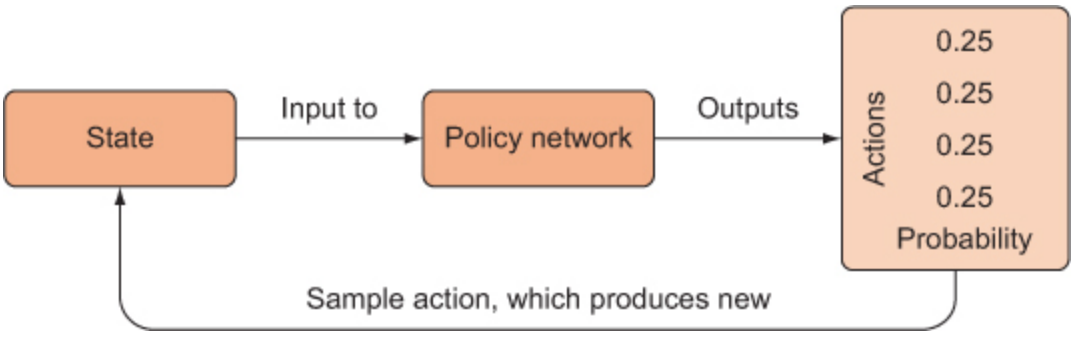
easy notation and discussion, it is standard to consider all the network parameters together as a single long vector that we denote θ (theta).

Whenever we run the policy network forward, the parameter vector θ is fixed; the variable is the data that gets fed into the policy network (i.e., the state). Hence, we denote the parameterized policy as π_θ. Whenever we want to indicate that some input to a function is fixed, we will include it as a subscript rather than as an explicit input like $\pi(x,\theta)$ where x is some input data (i.e., the state of the game). Notations like $\pi(x,\theta)$ suggest θ is a variable that changes along with x, whereas π_θ indicates that θ is a fixed parameter of the function.

Let's say we give our initially untrained policy network π_θ some initial game state for Gridworld, denoted s, and run it forward by computing $\pi_\theta(s)$. It returns a probability distribution over the four possible actions, such as [0.25, 0.25, 0.25, 0.25]. We sample from this distribution, and since it's a uniform distribution, we end up taking a random action (figure 4.5). We continue to take actions by sampling from the produced action distribution until we reach the end of the episode.

Figure 4.5. The general overview of policy gradients for an environment with four possible discrete actions. First we input the state to the policy network, which produces a probability distribution over the actions, and then we sample from this distribution to take an action, which produces a new state.



Remember, some games like Gridworld are episodic, meaning that there is a well-defined start and end point to an episode of the game. In Gridworld, we start the game in some initial state and play until we either hit the pit, land on the goal, or take too many moves. So an episode is a sequence of